

on the basis of quoted spread that the 10 mils access fee will lead to increased trading costs and lower liquidity.

Crucially, the reasoning above applies only to a stock with an economic spread of greater than the tick.¹⁴⁶¹ When the economic spread is less than a tick, rebates funded by fees result in a pricing distortion, as section VII.B.3.b explains. The price at which liquidity providers would be willing to offer liquidity is less than one tick in the presence of the rebate. However, the tick forms a binding price floor, leading to an oversupply of liquidity. Specifically, the set price of liquidity results in economic rents that accrue to some at the expense of others, in this case to those able to get to the front of the queue the fastest. For these stocks, lowering the access fee will better equate supply and demand and lower transaction costs for investors broadly.

One commenter discussed the introduction of a rebate for sub-dollar trades on MEMX and MIAX.¹⁴⁶² The rebate, when introduced, was initially set to 0.3% of the dollar value of the trade,¹⁴⁶³ and was reduced to 0.05% several days later. The commenter presents empirical results indicating that the effective spread fell from approximately 0.4% to 0.25% of dollar value after the introduction of the rebate,¹⁴⁶⁴ and almost completely reversed back to 0.4% days later when

¹⁴⁶¹ See section VII.B.1 for the definition of the economic spread.

¹⁴⁶² See Nasdaq Letter II at 5-6 (stating “spreads would widen if access fees were to become inadequate to fund rebates to market makers and other participants that provide displayed liquidity to the markets. This widening would likely be significant, as the data below suggests. It shows that in early December 2020, when MIAX and MEMX first introduced rebates for sub-dollar stocks, spreads for such stocks fell dramatically, but when MIAX and MEMX then slashed rebates soon thereafter, spreads reverted to their prior levels.”).

¹⁴⁶³ That is, the rebate for sub-dollar trades was initially set equal to the access fee cap for sub-dollar trades.

¹⁴⁶⁴ See Nasdaq Letter II at 6. The effective spread is calculated as the signed difference between the execution price of a trade and the prevailing midpoint (i.e., the execution price minus the midpoint for buy orders and the midpoint minus the execution price for sell orders); the commenter then divides this by the midpoint price to arrive at the effective spread as a percentage of the price (mirroring the fact that the rebate is paid as a percentage of the execution price). The effective spread differs from the quoted half-spread because a trade may receive price improvement—that is, the trade may execute at a better price than the best quote, so that the effective spread is lower than the quoted half-spread—or a large trade may execute against

the rebate was reduced to 0.05%.¹⁴⁶⁵ In short, a rebate of 0.3% of dollar value led to a reduction in effective spreads of 0.15% of dollar value. The commenter's empirical result is consistent with the model presented in Figure 1. The model presented in Figure 1 predicts that a rebate will cause the liquidity supply curve to shift by the amount of the rebate—liquidity suppliers are willing to offer liquidity at a lower price on account of the rebate. In contrast to Panel B of Figure 1, however, the commenter's example does not include an increase in the access fee to fund the rebate; therefore, the commenter's example can be modelled by recreating Panel B without the shift in the demand curve—i.e., the introduction of the rebate will cause the equilibrium outcome to shift from the point where the dotted lines intersect to the point where the solid supply curve intersects with the dotted demand curve. The model therefore has multiple empirical predictions for the commenter's example: when the rebate is introduced, without a similar increase in access fees, the model predicts that spreads will fall and the equilibrium amount of liquidity transacted will rise; when the rebate is rolled back, the model predicts that

multiple levels of the order book. Both the effective spread and the quoted spread are measures of liquidity, but the quoted half-spread measures the prospective cost of trading immediately at the best available prices while the effective spread measures the ex-post cost of trading immediately (accounting for hidden orders and other sources of price improvement not known ex-ante, as well as order size). Additionally, because the effective spread measures the ex-post cost of trading immediately, it can only be calculated in the presence of a trade. Therefore, the effective spread is typically calculated by taking a weighted average of the effective spread across transactions—the commenter, for example, weighted the effective spread by the notional amount of each transaction. The quoted spread can be averaged over time—as with the TWAQS—because it is an ex-ante measure.

¹⁴⁶⁵

The commenter's results can be exhibited with a numerical example. Suppose in the absence of rebates a stock trades with a best offer of \$0.52 and a best bid of \$0.48, yielding a midpoint of \$0.50. A liquidity supplier at the offer would therefore receive proceeds of \$0.52 when their offer is executed against. The effective spread in the commenter's example would be calculated as the distance between the execution price and the midpoint, divided by the midpoint: $(\$0.52 - \$0.50)/\$0.50 = 4\%$. Now suppose that a rebate of 0.3% is offered by the exchange. In the commenter's analysis, this reduces the effective spread by 0.15% to 3.85% (from 4%). This implies that the offer price would shrink from \$0.52 to approximately \$0.51925 (keeping the midpoint constant at \$0.50 and using the fact that the effective spread must equal the difference in the ask and the midpoint, divided by the midpoint so that $3.85\% = (\$0.51925 - \$0.50)/\$0.50$). The liquidity provider would therefore earn \$0.51925 plus the rebate of 0.30% for a total proceed of \$0.5208 $(\$0.51925 + 0.003 * \$0.51925)$.

spreads will rise and the equilibrium amount of liquidity transacted will fall. The model's predictions on spreads are borne out by the commenter's data—spreads fell 0.15% when the rebate of 0.3% was in place, and spreads reverted when the rebate was rolled back.¹⁴⁶⁶ The model's prediction on the quantity of liquidity transacted are also borne out by Commission analysis—when the 0.3% rebate was in place, the dollar-volume of sub-dollar trades increased by a factor of three.¹⁴⁶⁷ In sum, the introduction of a rebate for sub-dollar trades on MEMX and MIAX resulted in a market reaction that is directionally consistent with the Commission's economic model presented in section VII.B.3.a and Figure 1. The large and abrupt tripling of trading volume is also consistent with concerns that rebates cause excessive intermediation.¹⁴⁶⁸

Some comments address the question of incentives for trading on exchanges. These commenters state that, as a result of the Commission's adoption of 10 mils versus 15 mils/30 mils alternative, the volume on lit exchanges will decline.¹⁴⁶⁹ Other commenters disagreed, stating that lower access fees could lead volume on exchanges to increase.¹⁴⁷⁰ However, the

¹⁴⁶⁶ The fact that spreads fell by less than the amount of the rebate indicates that rebates do not generally lower trading costs beyond the cost of funding the rebate; this is contrary to one commenter's statement that, "any cost savings non-retail investor participants realize from a reduction in the access fee cap are likely to be more than consumed by the rising frictional costs...associated with wider spreads." See Cboe Letter IV at 5, and further discussion surrounding infra note 1492.

¹⁴⁶⁷ In the week of Nov. 23, 2020, there was daily trade volume at sub-dollar execution prices of approximately \$330 million; the figure was \$383 million in the week of Dec. 7. The intervening week—the week of MEMX's 0.3% rebate for sub-dollar executions—saw \$1,025 million in daily trade volume at sub-dollar prices. The calculations are constructed using all normal trades that execute during normal trading hours from TAQ. Following the methodology in Nasdaq Letter II at 6, the calculations for the week of Nov. 30 exclude Nov. 30 and Dec. 4.

¹⁴⁶⁸ See supra note 1005, and see also the Proposing Release, supra note 11, at 80292.

¹⁴⁶⁹ See, e.g., Cboe Letter II at 8-9, Cboe Letter IV at 3-5, Nasdaq Letter I at 22-23, Nasdaq Letter II at 4, Nasdaq Letter IV at 9, and Nasdaq Letter V at 2, predicting that a reduction in rebates will increase segmentation and may make ATs and single-dealer platforms more attractive. See also infra note 1761 and surrounding text.

¹⁴⁷⁰ See Better Markets Letter I at 15 stating that "A reduction in access fees will impose lower costs on investors, removing a disincentive for trading on exchanges." Healthy Markets Letter I at 22, stating "Brokers' avoidance of these [access] fees is a significant contributor for brokers often choosing to

above analysis indicates that liquidity providers would not be deterred from quoting on exchange because they could widen the quote, thereby receiving the same economic profit as they received with the rebate. Liquidity demanders would not be worse off because the reduction in access fee would offset, or, in the case of stocks with an economic spread of less than a tick, more than offset, the increase in spread.

Commenters specifically stated that posted quotes on exchange face the risk of adverse selection. They state that a premium is necessary to compete with the off-exchange market, and that the rebate provides that premium. As other commenters state, this does not take into account the access fee, which (all else equal) discourages liquidity takers from accessing exchanges. Moreover, a premium can come in the form of the spread as opposed to a rebate. While the order protection rule requires that trading centers enforce policies and procedures that are reasonably designed to prevent trades from being executed at a price worse than the protected quote, nothing prevents off-exchange non-displayed liquidity being at a better price for the liquidity taker and worse price for the maker, and indeed that happens under the current fee/rebate structure. Rather than moving liquidity off-exchange, liquidity providers could widen the difference between on- and off-exchange quotes, leaving the underlying economic tradeoff the same.

The above analysis shows that the same opportunities that are available on-exchange in today's environment are still expected to be available with the adoption of these amendments, even under the lower access fee cap (indeed, these opportunities are expected to improve due to

internalize or first route to ATSs or OTC market makers, rather than to exchanges", IEX Letter I at 26 (stating "A substantial reduction in the access fees will be impactful for those investors and is likely to increase their willingness to trade on exchanges....The result can be an increase in the use of displayed exchange trading and an improvement in the price discovery function of the market, with broad benefits extending beyond trading on exchanges themselves"), See also IEX Letter IV at 18-19, BMO Letter at 3, and Themis Letter at 7-8.

the amendments to Rule 612). However, commenters state that the off-exchange environment may change due to the amendments.¹⁴⁷¹ These commenters raise concern regarding the amount of liquidity that is displayed versus non-displayed. One commenter stated that wider quoted spreads on exchange increase the range of prices at which trades execute off-exchange.¹⁴⁷² Because Rule 611 generally requires that off-exchange trades execute within the NBBO, as on-exchange spreads widen, a liquidity provider, now facing a wider NBBO, would be able to offer a wider spread off-exchange than that liquidity provider could do now. The commenter appears concerned that this ability to offer a wider spread off-exchange than previously will attract liquidity to off-exchange, and more specifically, non-displayed venues.

However, while liquidity providers would have the ability to offer wider spreads off-exchange than prior to the amendments, they would not necessarily have the incentive to do so. For while wider spreads would mean greater profits for the liquidity provider, that is only the case if their orders are filled. As stated by a commenter, off-exchange liquidity would still need to compete with on-exchange liquidity, and that on-exchange liquidity is now less expensive to access due to a lower access fee cap.¹⁴⁷³ If the spread off-exchange were to widen, non-displayed off-exchange quotes would be unlikely to attract liquidity takers. Therefore, there is not an incentive for liquidity providers to migrate off-exchange due to wider spreads on exchange. To summarize, spreads may widen on-exchange increasing pricing flexibility off-exchange, even so

¹⁴⁷¹ See *supra* note 1479 and surrounding discussion.

¹⁴⁷² See Cboe Letter III at 5: “Wider spreads are likely to most benefit wholesale broker-dealers, that may be able to offer more levels of price improvement, but at the expense of increased frictional costs for investors.”

¹⁴⁷³ See IEX Letter IV at 23: “The fact that exchanges use rebates to draw orders from other exchanges says nothing about the ability of exchanges to attract more orders that now go to off-exchange venues by using lower access fees and offering better execution quality.”

exchanges are not expected to lose volume due to the reduction in the access fee cap through this mechanism.

One commenter stated that volatility may increase due to the wider quoted spread when the access fee reduction causes a reduction in rebates.¹⁴⁷⁴ The Commission acknowledges that wider spreads definitionally imply a greater difference between the bid and the ask. However, spreads that better reflect the true underlying cost of liquidity are more efficient than spreads that mask this cost.

One commenter stated that a reduction in rebates will lead to more off-exchange trading, which in turn will cause the NBBO to widen, and result in worse execution for off-exchange trading, because of the way some off-exchange trading uses the NBBO as “a reference price for benchmark pricing and other risk functions.”¹⁴⁷⁵ First, the Commission describes above why the adopted amendments will not result in a large amount of trading moving off-exchange. Furthermore, while the Commission does expect the quoted spread, and therefore the NBBO, to widen, we disagree that this will result in worsening off-exchange executions. This commenter provided two examples of situations in which off-exchange executions might worsen. The first is an ATS that provides execution mechanisms based on the NBBO.¹⁴⁷⁶ As explained in section VII.B.3.a, the reduction in access fees and corresponding reduction in rebates will not change the net spread on exchange. This means the cost of liquidity will not materially change. There is no

¹⁴⁷⁴ See Goldman Sachs Letter at 8.

¹⁴⁷⁵ See Cboe Letter IV at 5: “...the NBBO is utilized by many market participants as a reference price for benchmark pricing and other risk functions. In addition, if on-exchange liquidity moves to off-exchange venues such as alternative trading systems, these trading centers commonly use the NBBO as a reference price for executing transactions, which will make transactions in off-exchange venues more expensive as well. Wider spreads are likely to most benefit wholesale broker-dealers, that may be able to offer more levels of price improvement, but at the expense of increased frictional costs for investors.”

¹⁴⁷⁶ See Cboe Letter IV at 5.

reason why ATSS that base execution prices off the NBBO cannot alter their pricing formulas to preserve the same execution prices (e.g., by executing inside the NBBO by a pre-determined amount). Indeed, a typical example of such matching mechanisms are mechanisms that match buy and sell orders at the midpoint, and this will not be impacted at all by a wider NBBO. The second example provided by the commenter was the case of retail wholesalers. The commenter states that these wholesalers “may be able to offer more levels of price improvement,” but this will come at the expense of increased costs of trading from wider spreads.¹⁴⁷⁷ The Commission again disagrees with this assertion. Because the cost of liquidity will be largely unchanged, the price improvement¹⁴⁷⁸ acknowledged by the commenter will be capable of offsetting the change in quoted spread.

One commenter stated that eliminating the access fee would cost retail investors as much as \$678 million per year.¹⁴⁷⁹ The commenter arrives at this estimate by using the BJZZ algorithm to identify retail trades from TAQ data,¹⁴⁸⁰ and computes the effective/quoted ratio (EQ ratio) for each trade.¹⁴⁸¹ They assume that spreads will widen for all trades by 60 mils in the absence of rebates. They then apply the observed EQ ratio to the hypothetical 60 mil wider spreads in the absence of rebates to compute hypothetical transaction costs for retail investors under a world without rebates.

¹⁴⁷⁷ See Cboe Letter IV at 5.

¹⁴⁷⁸ Retail wholesalers frequently offer “price improvement” on orders they receive, where they execute the order on a principal basis at a price better than the NBBO.

¹⁴⁷⁹ See Nasdaq Letter II at 6.

¹⁴⁸⁰ “BJZZ” refers to the algorithm designed by Boehmer Ekkehart, Charles M. Jones, Xiaoyan Zhang, and Xinran Zhang. See Boehmer Ekkehart, et al., Tracking Retail Investor Activity, 76 J. FIN, 2249 (2021).

¹⁴⁸¹ The effective to quoted spread ratio computed as follows $EQratio = \frac{|Execution\ Price - NBBO\ midpoint|}{NBBO\ Half\ Spread}$. It measures how close to the NBBO or NBBO midpoint a trade executes at. A trade executing at the midpoint would have an EQ ratio of 0, while a trade that executes at the NBBO would have an EQ ratio of 1.

The Commission disagrees with the commenter's assertion that retail traders will receive worse execution due to the reduced access fee cap. As this section describes, quoted spreads for stocks trading with more than one tick intra-spread are expected to widen on average by about 40 mils. However, the commenter's analysis relies on the assumption that the EQ ratio for retail order executions will remain constant.¹⁴⁸² The commenter provided no evidence to support this assumption. This assumption is important because economically what matters is not the distance of the trade price from the NBBO, but rather the distance of the trade price from the midpoint – the effective spread. The effective spread covers the costs associated with providing liquidity as well as provides the liquidity provider's profits. Assuming a constant EQ ratio in the commenter's analysis implies that wholesalers internalize retail orders at prices that are farther from the midpoint, and thus the wholesaler will earn more money without providing any additional benefit to retail traders or their broker-dealers.¹⁴⁸³ Wholesalers are subject to competitive forces that apply at the level of average execution quality, and it is unlikely that market forces would allow such excess profits to wholesalers for no additional benefit to persist.¹⁴⁸⁴ It also seems unlikely that the EQ ratio would change mechanically with the NBBO

¹⁴⁸² The commenter's methodology is also flawed because the BJZZ algorithm they employ to identify retail trades has been shown in recent research as not being a very accurate measure of retail trading volume, See Brad M. Barber, Xing Huang, Philippe Jorion, Terrance Odean, & Christopher Schwarz, A(sub)penny For Your Thoughts: Tracking Retail Investor Activity in TAQ (working paper, Aug. 14, 2023), available at <https://ssrn.com/abstract=4202874> (retrieved from SSRN Elsevier database. If the algorithm does not reliably identify retail trades then it is unclear what can actually be learned about retail trading volume from the exercise.

¹⁴⁸³ The effective spread is defined as the signed difference between an order's execution price and the midpoint of the quoted spread; the larger the difference the less competitive the executed price is relative to the midpoint. Because the EQ ratio is equal to the effective spread divided by the quoted spread, the effective spread would have to increase at the same scale by which the quoted spread widens in order to keep the ratio constant.

¹⁴⁸⁴ So long as there is some degree of competition, this argument would hold. A market that is more competitive may have retail effective spreads that would be lower, but in either case a change in the NBBO, with no other changes to wholesaler costs or competition would not be expected to change wholesaler profits.

because if the NBBO itself were the primary determinate of the price level at which retail trades were internalized, and wholesalers were free to choose any price level within the NBBO, then wholesalers would routinely internalize orders at or near the NBBO implying an EQ ratio for retail trades of near 1. This is not the case. Wholesalers currently internalize retail orders at prices that are significantly inside of the NBBO (i.e., EQ ratios significantly less than 1) suggesting that other factors besides the NBBO itself, such as distance from the midpoint, determine the transaction price of retail orders that are internalized by wholesalers.¹⁴⁸⁵ These price levels will still be feasible for wholesalers under the amendments, and so even with a wider NBBO, wholesalers are likely to transact retail orders at similar price levels under the amendments as they are today.

Put another way, economically there is no reason to assume that relaxing a non-binding constraint, in this case widening the quoted NBBO, would have an effect on existing equilibrium behavior. The NBBO does not constitute a binding constraint for wholesale execution of many retail trades.

In cases where the NBBO does constitute a binding constraint, there are two important missing pieces from the commenter's analysis. The first is that, for on-exchange execution, the wholesaler will pay a lower access fee. Assuming (as the commenter's analysis implicitly does) that the wholesaler does not pass on these lower fees at least in part to some investors assumes a lack of competitive dynamics in the retail execution market. Second, the commenter's methodology fails to take into account the expected reduction in quoted spreads for some stocks

¹⁴⁸⁵ See Citadel Letter I at 33 showing EQ ratios ranging from .27 for small retail orders to .88 for very large orders, CCMR Letter at 35 showing average EQ ratios around .5 for the top three wholesalers and Charles Schwab, U.S. Equity Market Structure: Order Routing Practices, Considerations, and Opportunities. (2022) ("Schwab 2022 Whitepaper") at 9, 16, available at <https://content.schwab.com/web/retail/public/about-schwab/Schwab-2022-order-routing-whitepaper.pdf> (showing its EQ ratio of .33).

due to the reduction in the tick size. In cases where the NBBO constitutes a binding constraint on wholesaler price improvement, then wholesalers will offer better execution on these stocks.

Moreover, there are cases in which current wholesale execution may fall between the spread under the new tick size, and the previous spread. In these cases, the new tick size creates a new binding constraint, leading to better execution for retail investors. So, if anything, the combined effect of the amendments could improve retail execution quality on average.

Additionally, as a general matter, some commenters stated that the lower access fee on exchanges will make exchanges a more attractive place to access liquidity.¹⁴⁸⁶ The Commission believes that the cost of accessing liquidity will decline for those stocks which continue to trade with a one-tick wide spread; the Commission, however, disagrees that the cost of accessing liquidity will change on average for other stocks.¹⁴⁸⁷

Some commenters stated concerns that less liquid stocks may be more susceptible to any negative effects on liquidity from a reduction in rebates.¹⁴⁸⁸ Other commenters suggested that a higher fee cap should be adopted for illiquid stocks.¹⁴⁸⁹ However, adopting a separate fee cap for illiquid stocks would introduce more complexity into the market. The Commission's response is the same as the broader concern regarding posted liquidity: spreads may widen but the cost of accessing and providing liquidity will on average not change.

¹⁴⁸⁶ See Healthy Market Letter at 23, BlackRock Letter at 11, BMO Letter at 3, and Themis Letter at 7-8.

¹⁴⁸⁷ For stocks which do not trade with a one-tick wide spread the cost of accessing liquidity for any one instance may be higher or lower with a reduced fee cap, however on average the cost of accessing liquidity is not expected to change for those stocks. See supra section VII.B.3.b for additional discussion.

¹⁴⁸⁸ See Nasdaq Letter II at 5-7: "This peril is particularly acute for thinly-traded securities." and Virtu at 10: "The reduced incentives for liquidity in thinly traded securities is especially concerning given how much liquidity improvements actually reduce an issuer's cost of capital and impact their ability to attract investors." See also Virtu Letter II at 10, Tastytrade Letter at 2.

¹⁴⁸⁹ See Citigroup Letter at 6, TRP Letter at 4-5.